Coffee Shop Sales & Inventory Management System using SQL

Mini Project Report

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1. Introduction

The Coffee Shop Sales & Inventory Management System is a SQL-based database project developed to manage coffee shop operations efficiently. The project focuses on handling customer orders, inventory tracking, staff management, and sales analysis.

This system helps store and analyze coffee shop data using SQL queries and relational database concepts.

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2. Objectives

- To manage coffee shop sales records
- To maintain inventory details
- To manage staff information
- To analyze customer orders
- To generate sales insights using SQL queries
- To understand relational database management systems

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3. Technologies Used

Technology	Purpose
MySQL	Database Management
MySQL Workbench	SQL Query Execution
CSV Files	Data Source
SQL	Data Analysis

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4. Database Name

```
Coffee_Shop_Sales
```

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5. Tables Used

Table Name	Description
staff	Employee details
orders	Customer order details
items	Coffee and menu items
shift	Shift information
users	User details
login_logs	Login information
organizations	Organization records
systems	System records
security_incidents	Security incident records

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6. Table Structure

Staff Table

Column Name	Data Type
staff_id	VARCHAR
first_name	VARCHAR
last_name	VARCHAR
position	VARCHAR
sal_per_hour	INT

Orders Table

Column Name	Data Type
row_id	INT
order_id	VARCHAR
created_at	DATETIME
item_id	VARCHAR
quantity	INT
cust_name	VARCHAR
in_or_out	VARCHAR

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7. SQL Queries Used

Query 1 — Most Ordered Items

```
SELECT
    item_id,
    COUNT(*) AS total_orders
FROM orders
GROUP BY item_id
ORDER BY total_orders DESC;
```

Purpose: Finds the most ordered coffee items.

Query 2 — Customer with Most Orders

```
SELECT
    cust_name,
    COUNT(order_id) AS total_orders
FROM orders
GROUP BY cust_name
ORDER BY total_orders DESC;
```

Purpose: Identifies the customers placing the highest number of orders.

Query 3— Orders Type Analysis

```
SELECT
    in_or_out,
    COUNT(*) AS total_orders
FROM orders
GROUP BY in_or_out;
```

Purpose: Analyzes dine-in and takeaway orders.

Query 4 — Staff Salary Analysis

```
SELECT
    position,
    AVG(sal_per_hour) AS avg_salary
FROM staff
GROUP BY position;
```

Purpose: Calculates average salary by position.

Query 5 — Daily Order Count

```
SELECT
    DATE(created_at) AS order_date,
    COUNT(*) AS total_orders
FROM orders


GROUP BY DATE(created_at)
ORDER BY order_date;
```

Purpose: Tracks daily order trends.

Query 6 — Top Customers

```
SELECT
    cust_name,
    SUM(quantity) AS total_items
FROM orders
GROUP BY cust_name
ORDER BY total_items DESC
LIMIT 5;
```

Purpose: Displays top customers based on quantity purchased.

Query 7— Ranking Customers using Window Function

```
SELECT
    cust_name,
    quantity,
    RANK() OVER(ORDER BY quantity DESC) AS ranking
FROM orders;
```

Purpose: Ranks customers based on order quantity.

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8. SQL Concepts Used

Concept	Usage
SELECT	Retrieve data
WHERE	Filter records
GROUP BY	Group records
ORDER BY	Sort results
Aggregate Functions	SUM, COUNT, AVG
LIMIT	Restrict rows
Window Functions	Ranking

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9. Advantages of the System

- Easy management of coffee shop operations
- Efficient sales tracking
- Inventory monitoring
- Better customer analysis
- Staff performance evaluation
- Faster data retrieval using SQL

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10. Conclusion

The Coffee Shop Sales & Inventory Management System successfully demonstrates the implementation of SQL concepts in a real-world business environment. The project helps in managing sales, inventory, and staff operations efficiently.

This project improved understanding of:

- Relational databases
- SQL query writing
- Data analysis
- Business reporting

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11. Future Enhancements

- Power BI dashboard integration
- Online ordering system
- Automated inventory alerts
- Customer loyalty tracking
- Mobile application integration